

StellarPGx: A Nextflow Pipeline for Calling Star Alleles in Cytochrome P450 Genes

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Bioinformatics pipelines for calling star alleles (haplotypes) in cytochrome P450 (CYP) genes are important for the implementation of precision medicine. Genotyping CYP genes using high throughput sequencing data is complicated, e.g., by being highly polymorphic, not to mention the structural variations especially in *CYP2D6*, *CYP2A6*, and *CYP2B6*. Genome graph-based variant detection approaches have been shown to be reliable for genotyping *HLA* alleles. However, their application to enhancing star allele calling in CYP genes has not been extensively explored. We present StellarPGx, a Nextflow pipeline for accurately genotyping CYP genes by combining genome graph-based variant detection, read coverage information from the original reference-based alignments, and combinatorial diplotype assignments. The implementation of StellarPGx using Nextflow facilitates its portability, reproducibility, and scalability on various user platforms. StellarPGx is currently able to genotype 12 important pharmacogenes belonging to the CYP1, 2, and 3 families. For purposes of validation, we use *CYP2D6* as a model gene owing to its high degree of polymorphisms (over 130 star alleles defined to date, including complex structural variants) and clinical importance. We applied StellarPGx and three existing callers to 109 whole genome sequenced samples for which the Genetic Testing Reference Material Coordination Program (GeT-RM) has recently provided consensus truth *CYP2D6* diplotypes. StellarPGx had the highest *CYP2D6* diplotype concordance (99%) with GeT-RM compared with Cyrius (98%), Aldy (82%), and Stargazer (84%). This exemplifies the high accuracy of StellarPGx and highlights its importance for both research and clinical pharmacogenomics applications. The StellarPGx pipeline is open-source and available from <https://github.com/SBIMB/StellarPGx>.

Study Highlights

WHAT IS THE CURRENT KNOWLEDGE ON THE TOPIC?

✓ As high throughput sequence data becomes increasingly available in clinical and research settings, bioinformatics tools for accurately calling star alleles in highly polymorphic pharmacogenes, via diverse approaches, are required to facilitate pharmacogenomics testing and allele characterization in various populations.

WHAT QUESTION DID THIS STUDY ADDRESS?

✓ Our study addresses the need to implement a pipeline for accurate, scalable, and reproducible genotyping of cytochrome P450 (CYP) genes using high throughput sequence data. In addition, our pipeline performs automated phenotype prediction for well-characterized CYP genes such as *CYP2D6*.

WHAT DOES THIS STUDY ADD TO OUR KNOWLEDGE?

✓ This study shows that StellarPGx can provide accurate, scalable, and reproducible CYP genotyping by combining pangenome-based variant detection, efficient combinatorial diplotype assignment, and read coverage information from high throughput sequence data.

HOW MIGHT THIS CHANGE CLINICAL PHARMACOLOGY OR TRANSLATIONAL SCIENCE?

✓ StellarPGx will facilitate CYP star allele calling in the clinical pharmacology and translational science communities. StellarPGx will also afford the ability to identify individuals with possible novel CYP alleles.

The human cytochrome P450 (CYP) gene family currently comprises 57 genes. Of these, 12 (belonging to the CYP1, 2, and 3 families) encode enzymes responsible for the bioactivation

or elimination of over 80% of commonly prescribed medications. These CYP genes are also known to be highly polymorphic.¹ Accordingly, they account for the highest number of

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pharmacogenetic biomarkers in drug labels according to the US Food and Drug Administration (FDA).² Genotyping CYP genes and subsequent phenotype prediction is therefore a fundamental part of precision medicine approaches as CYP allele information is used for adjusting therapy, thus countering drug inefficacy and adverse drug reactions.^{3–5}

CYP gene haplotypes—single nucleotide polymorphisms, small insertions and deletions (indels), and/or structural variations that are inherited together—are commonly denoted using the star (*) allele nomenclature to aid in interpretation.⁶ We refer to single nucleotide polymorphisms and indels collectively as small nucleotide variations (SNVs) for the rest of this manuscript. The metabolizer activity of star alleles for most CYP genes ranges from nonfunctional to increased function, making it possible to predict individual phenotypes from diplotypes via activity score systems.^{6–9} This makes accurate genotyping of CYP genes very useful and imperative. Notably, the Pharmacogene Variation Consortium (PharmVar)^{10,11} and the Pharmacogenomics Knowledgebase (PharmGKB)¹² provide resourceful, curated catalogues of the haplotype information for CYP genes (and other pharmacogenes) as well as links to the related clinical annotations, respectively.

The ever-decreasing cost and increasing ubiquity of high-throughput sequencing (HTS) provide the opportunity for clinical-grade whole genome sequencing (WGS) and extensive characterization of CYP allele distribution especially in previously understudied populations. However, single molecule long-read technologies, which would make haplotype detection straightforward^{13–16} are more expensive and less accessible compared with short-read technologies such as Illumina.¹⁷ Notably, calling star alleles in CYP genes using short-read next-generation sequencing (NGS) data is challenging especially for genes which have neighboring pseudogenes (e.g., *CYP2D6*, *CYP2B6*, and *CYP2A6*) or isoforms (e.g., *CYP2C19* and *CYP2C9*) with high sequence similarity, hence causing a number of structural rearrangements, sequencing problems and subsequent read misalignments.^{18,19}

The current state-of-the-art bioinformatics algorithms for calling star alleles in CYP and/or other pharmacogenes from short-read NGS data include Astrolabe (formerly Constellation),²⁰ Aldy,²¹ Stargazer,^{22,23} PharmCAT,²⁴ and Cyrius.²⁵ These algorithms mainly use binary alignment map (BAM) files and/or Variant Call Format (VCF) files as input to call SNV-defined alleles as well as structural variants. A notable commonality among these existing tools is that they all rely on variant detection based on reads mapped to a standard reference genome^{26,27} (Genome Reference Consortium Human Build 37 (GRCh37) or GRCh38) as a fundamental step in their diplotype calling. Although this approach has facilitated variant calling from HTS data over the years, it has been shown to have limitations such as reference allele bias and read misalignment around indels.^{28–30} Aligning reads to a reference genome without awareness of variation could, therefore, lead to erroneous variant calling and subsequently inaccurate star allele calling.

There is a growing paradigm shift towards using pangenomes to counter the limitations of reference-based alignments. A pangenome is a graph structure characterized by having a linear reference genome augmented with prior variant information such that each

haplotype is represented as a path in the graph.^{28–30} This facilitates more accurate read alignment as well as simultaneous genotyping of variants present in the graph.³¹ Furthermore, novel variants can still be accurately called and used to update the graph. Given the high degree of polymorphisms and complex variation in most CYP genes, they are an excellent application for pangenome-based alignment and genotyping approaches. High genotyping accuracy, courtesy of genome graphs, has already been shown for similarly complex genomic regions such as the human leukocyte antigen (*HLA*) complex^{32,33} and short tandem repeat regions.³⁴

Here we present StellarPGx, a highly efficient hybrid pipeline that combines genome graph-based variant detection, read coverage information from the original HTS data alignments, and combinatorial diplotype assignment for genotyping CYP genes in a reproducible manner. StellarPGx is implemented using Nextflow,³⁵ an *in silico* workflow management system that facilitates parallelization, scalability, reproducibility, and portability of pipelines via Docker (<https://www.docker.com>) and Singularity³⁶ technologies. Docker and Singularity allow for the easy distribution of all the software dependencies for StellarPGx in a containerized system, thus overcoming reproducibility issues caused by the diversity in user computational frameworks.

We use *CYP2D6* as a model gene for validating StellarPGx using WGS data from 109 DNA samples that have been characterized as part of the Centers for Disease Control and Prevention's (CDC's) Genetic Testing Reference Materials Coordination Program (GeT-RM).³⁷ *CYP2D6* provides the ideal use case for StellarPGx as over 130 star alleles have been catalogued by PharmVar 10, including complex structural rearrangements with the neighboring nonfunctional paralog, *CYP2D7*. GeT-RM recently recharacterized 179 samples to resolve truth diplotypes for *CYP2D6* based on consensus results from various laboratories that used orthogonal genotyping techniques such as XL-PCR, Sanger sequencing, PacBio sequencing, and TaqMan copy number variation analysis.³⁷ One hundred and nine of the GeT-RM samples have publicly available short-read WGS data covering over 40 *CYP2D6* star alleles, which we used for the validation of StellarPGx and comparison with Aldy and Stargazer, which we had found to be the best-performing tools according to our previous benchmark,³⁸ as well as Cyrius 25 which was more recently developed.

METHODS

Implementation

We developed StellarPGx using Nextflow³⁵, a domain-specific language for creating scalable, highly parallelizable and portable pipelines with components that can be written in various programming languages. StellarPGx is containerized using Docker and Singularity for easy distribution of all the required software dependencies, thus facilitating workflow reproducibility on user systems. The major steps in the StellarPGx pipeline are summarized in **Figure 1** and described below using *CYP2D6* as a model gene. The StellarPGx version described here is currently able to identify *CYP2D6*1* through *CYP2D6*141*.

Input data and graph-based variant calling

StellarPGx takes BAM or CRAM files as input since they are the most common formats for storing and sharing WGS data that can be used for both SNV calling and copy number determination. StellarPGx supports alignments to both GRCh37 and GRCh38.

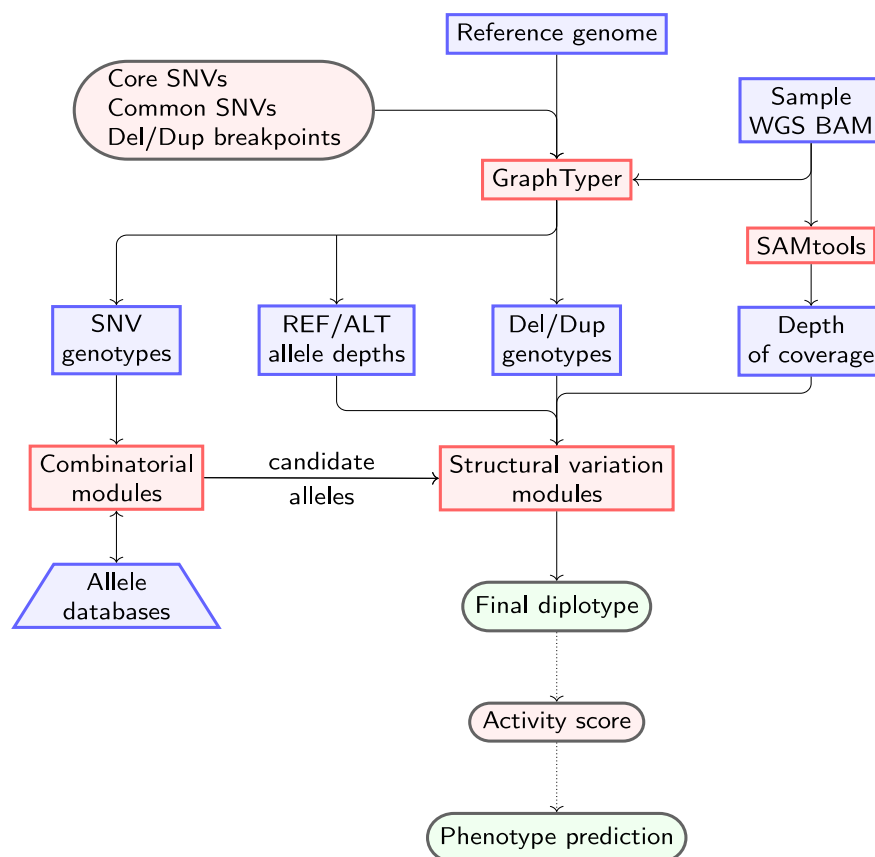


Figure 1 Overall summary of the StellarPGx workflow. The supported input in the version described here is a WGS (whole genome sequence) BAM (binary alignment map) file. Common and allele-defining SNVs (small nucleotide variations) as well as Del (gene deletion) and Dup (duplication) breakpoints are supplied in Variant Call Format for GraphTyper's pangenome construction. Copy number sentinel regions (target gene and control loci) are supplied in a BED (browser extensible data) file for read depth computation. StellarPGx's Nextflow processes then use the output from GraphTyper and SAMtools bedcov as well as input from allele/suballele combination databases to call sample diploypes for a target cytochrome P450 (CYP) gene via Python 3.8 custom modules. Subsequent steps include determining the AS (activity score) based on the genotype followed by phenotype prediction depending on the AS group. These last two steps have not yet been implemented for all the supported CYP genes as consensus AS and/or phenotype assignment recommendations are available for only a few genes such as *CYP2D6*, *CYP2B6*, *CYP2C19*, and *CYP2C9*. ALT, alternate allele; REF, reference allele.

Typical steps in pangenome-based variant detection (via the variation graph route rather than assembly) include genome graph construction using a reference genome and input variants (usually common variants in a population), graph indexing, alignment of sequence reads to the pangenome, and subsequent variant calling.²⁸ Rather than reinventing algorithms for these initial steps, StellarPGx uses GraphTyper.^{33,39} GraphTyper is able to retrieve sequence reads from a defined region in a BAM/CRAM file and realign them to a variant-aware graph (pangenome), thus supporting SNV and structural variant genotyping based on the paths the reads align to and alignment coverage or breakpoint genotypes, respectively. StellarPGx uses GraphTyper to realign all reads from *CYP2D6* and *CYP2D7* to a genome graph with known *CYP2D6* variants, and simultaneously return variant genotypes as well as allele depth information in VCF format for each sample (see **Supplementary Materials S1** for an illustration of this process). Importantly, GraphTyper output also comprises novel sample variants.

The default prior variants included in StellarPGx for constructing the pangenome include all the allele-defining *CYP2D6* SNVs, as well as variants with a global frequency above 1% according to either the 1000 Genomes Project⁴⁰ or gnomAD⁴¹ databases. For genotyping the *CYP2D6* full gene deletion and gene duplication via GraphTyper, StellarPGx uses a VCF with predefined imprecise breakpoints (see **Table S1**) for graph construction.

Star allele calling

StellarPGx's SNV-defined allele caller is written in Python 3.8 (<https://docs.python.org/3.8/>). It identifies known allele-defining variants from the VCF produced by GraphTyper and matches them to a plain-text StellarPGx database developed from PharmVar's star allele definitions.¹⁰ The StellarPGx database contains all possible combinations of known SNV-defined alleles for *CYP2D6* and other supported CYP genes. For most cases, a combination of core SNVs corresponds to one diplotype. In such a scenario, this diplotype is considered to be the candidate diplotype in the sample. For the cases where the presence of a particular combination of core SNVs matches more than one candidate diplotype (typically two or three), we query a StellarPGx subdatabase with suballele combinations to find the candidate suballele-resolved combination that best explains all the SNV genotypes in the sample. This two-step process helps identify star alleles faster especially for cases where there is only one candidate diplotype. However, there are some candidate diploypes which produce exactly the same variant combinations even when subvariants are considered. For example *CYP2D6**103.001/*3.001 and *CYP2D6**102.001/*3.002 produce a VCF with exactly the same variants as well as their zygosity (see **Supplementary Materials S1**). StellarPGx calls *CYP2D6**102/*3 or *103/*3 for a sample with such a combination. Such cases arise mostly due to alleles with a "limited" PharmVar level of evidence,¹⁰

meaning that their definitions are based on limited information such as exon-only sequencing as opposed to full gene resequencing.

The next step in the star allele calling process is discriminating structural variant alleles, if present. Due to the many *CYP2D6* structural variant rearrangements,¹⁰ we integrate read coverage data from the original reference-based WGS BAM/CRAM file as well to support calling of complex *CYP2D6* structural rearrangements with *CYP2D7*. We use SAMtools⁴² to compute the depth of coverage for predefined *CYP2D6*, *CYP2D7*, and control gene regions as well as exons/introns affected by known hybrid rearrangements. For example, exon 9 is a priority as there are a number of *CYP2D6-2D7* alleles with an exon 9 conversion including *CYP2D6*36*, **57*, and **83*.¹⁰ Details for the GRCh37 and GRCh38 regions used in StellarPGx's structural variant calling can be found in the StellarPGx project repository (<https://github.com/SBIMB/StellarPGx>). Vitamin D Receptor (*VDR*) and Epidermal Growth Factor Receptor (*EGFR*) genes have been shown to be effective as control genes in star allele calling²³ and they have relatively low copy number variation frequencies;^{43–45} hence we use the average coverage of both genes as a control for copy number. The implementation of StellarPGx in Nextflow allows for this depth-of-coverage computation to be performed in parallel with the aforementioned pangenome-based variant calling.

We determine the total copy number of *CYP2D6* alleles in the sample by comparing the read coverage for *CYP2D6* and its flanking regions with the read coverage for the control loci. If a copy number of 2 is ascertained, StellarPGx reports the candidate SNV diplotypes as the final result except for cases where we check for presence of possible singleton hybrids. For example, we use individual algorithms to check whether *CYP2D6*36* (exon 9 conversion) is present instead of the initially called *CYP2D6*10*. If a copy number of 1 is detected, StellarPGx replaces the homozygous diploidy assignments from the SNV information with a *CYP2D6*5* <allele> assignment* (**5* represents the *CYP2D6* gene deletion allele). For example, a sample with *CYP2D6*5*17* would initially be genotyped as *CYP2D6*17*17* based on SNVs alone. After running the SV caller algorithm, StellarPGx would then report the final diploidy as *CYP2D6*5*17*. The detection of copy number 0 would default to a *CYP2D6*5*5* diploidy call.

On the other hand, if a copy number of 3 or more is detected, StellarPGx uses the allele depth ratios (from GraphTyper VCF output) for each core SNV to decipher which of the two alleles in a sample is duplicated/multiplied. For duplications involving tandem rearrangements, StellarPGx runs hybrid caller modules to discriminate the hybrid copies based on information from PharmVar structural variant definitions.¹⁰ For example, let us consider the *CYP2D6*68*4* tandem (found frequently in Europeans) which comprises the nonfunctional **4* allele and the *CYP2D6-2D7* hybrid, i.e., *CYP2D6*68* (intron 1 switch region). StellarPGx ascertains the *CYP2D6*68*4* tandem in a sample with the *CYP2D6*41*68*4* diploidy by comparing the coverage from the promoter region to intron 1 vs. the coverage from intron 1 to exon 9 as well as the allele depth (and zygosity) for 100C > T (found in *CYP2D6*68* and most *CYP2D6*4* alleles) vs. that for 1847G > A (**4*) (see **Figure 2b**). Other significantly frequent tandems include *CYP2D6*36*10*, **36*2*, and **36*2*10*, which can be discriminated from *CYP2D6*10*2* and **10*3* by interrogating the *CYP2D6* exon 9 coverage (see **Figure 2c**). In some cases (such as *CYP2D6*13* hybrid alleles—see **Figure 2d**) the coverage of *CYP2D7* regions is also considered to enhance StellarPGx's recall. More details about the modules for calling the wide array of CYP structural variants (especially for *CYP2D6*) can be found in the StellarPGx project repository (<https://github.com/SBIMB/StellarPGx>).

StellarPGx output

The StellarPGx output file contains the final diploidy call and phenotype assignment (where applicable) of the target CYP gene as well as a breakdown of the variants and copy number information used for diploidy assignment. Other candidate solutions that were considered by the

caller (if applicable) are also provided as this information could be useful especially when performing ensemble star allele calling (using at least three tools to enhance accuracy)³⁸. Example output files are provided in **Supplementary Materials S1**.

For cases where StellarPGx detects a novel allele or suballele, it will indicate "Possible novel major allele or suballele present: Interpret with caution; Experimental validation and expert review through PharmVar is recommended" in the output (see **Supplementary Materials S1**). Novel allele discovery is more or less straightforward for samples with novel combinations of existing core variants. However, detecting novel alleles when completely novel core variants are present in a sample is more challenging. The current strategy is to include novel (i.e., not yet in the PharmVar database) missense, stop loss, stop gain, and frameshift CYP variants in the StellarPGx database so these can be flagged by StellarPGx when they appear in the GraphTyper-generated VCF. These variants include SNVs identified from large-scale genomics projects such as the 1000 Genomes Project⁴⁰ and gnomAD⁴¹, as well as possible missense variants that could occur in the target CYP genes even though they haven't been identified in previous studies. However, we are yet to include support for calling novel structural variants. The structural variants that are currently supported by StellarPGx are those defined by PharmVar as per the v4.2.4 update.¹⁰ We strongly recommend that users liaise with PharmVar¹⁰ for information about experimental validation of possible novel star alleles and/or their designation if any of these cases arise in clinical or research settings.

Validation

In order to validate the StellarPGx pipeline, we performed *CYP2D6* star allele calling on publicly available high coverage WGS data for 109 ethnically diverse reference samples from the GeT-RM project. This includes 75 samples from the previous GeT-RM work⁴⁶ and 34 of the 42 new samples added in the latest GeT-RM study³⁷, which was focused on recharacterization of *CYP2D6* alleles for 179 DNA samples (not all have WGS data publicly available) derived from Coriell cell lines. (See **Table S2** for details about the data sources.)

We compared *CYP2D6* star allele calls and phenotype assignments by StellarPGx (version described here covers *CYP2D6*1*141*) with the GeT-RM consensus reference materials (cover *CYP2D6*1*114*), Aldy v2.2.6 (covers *CYP2D6*1*114*), Stargazer v1.0.8 (covers *CYP2D6*1*138*), and Cyrius v1.1 (covers *CYP2D6*1*139*). For Aldy and Stargazer, we performed read realignment (*CYP2D* and control regions) for 34 of the samples from GRCh38 to GRCh37 using BWA-MEM.²⁶ We used the *VDR* gene as the control region for the Stargazer run. The control regions for Aldy and Cyrius are predetermined within their algorithms.

Furthermore, we genotyped 11 other CYP genes (including *CYP2A6*, *CYP2B6*, *CYP2C19*, *CYP2C9*, *CYP2C8*, *CYP3A4*, *CYP3A5*, *CYP2A1*, *CYP1A2*, *CYP2E1*, and *CYP4F2*) using 75 WGS samples corresponding to reference materials from the GeT-RM study by Pratt *et al.*,⁴⁶ to demonstrate the utility of StellarPGx in characterizing the diverse CYP allele landscape. However, for some genes, not all the samples had corresponding GeT-RM consensus calls reported. Therefore, StellarPGx's concordance was determined using the samples for which the GeT-RM consensus calls were available.

RESULTS

Comparison of StellarPGx with other callers based on *CYP2D6* diplotyping

We used published consensus *CYP2D6* diplotypes for the 109 WGS samples corresponding to the GeT-RM reference materials as the truth diploidy calls 38 except for sample NA18519. GeT-RM reported the consensus *CYP2D6* call for NA18519 as **1*29*. However, this sample's reference diploidy was based on

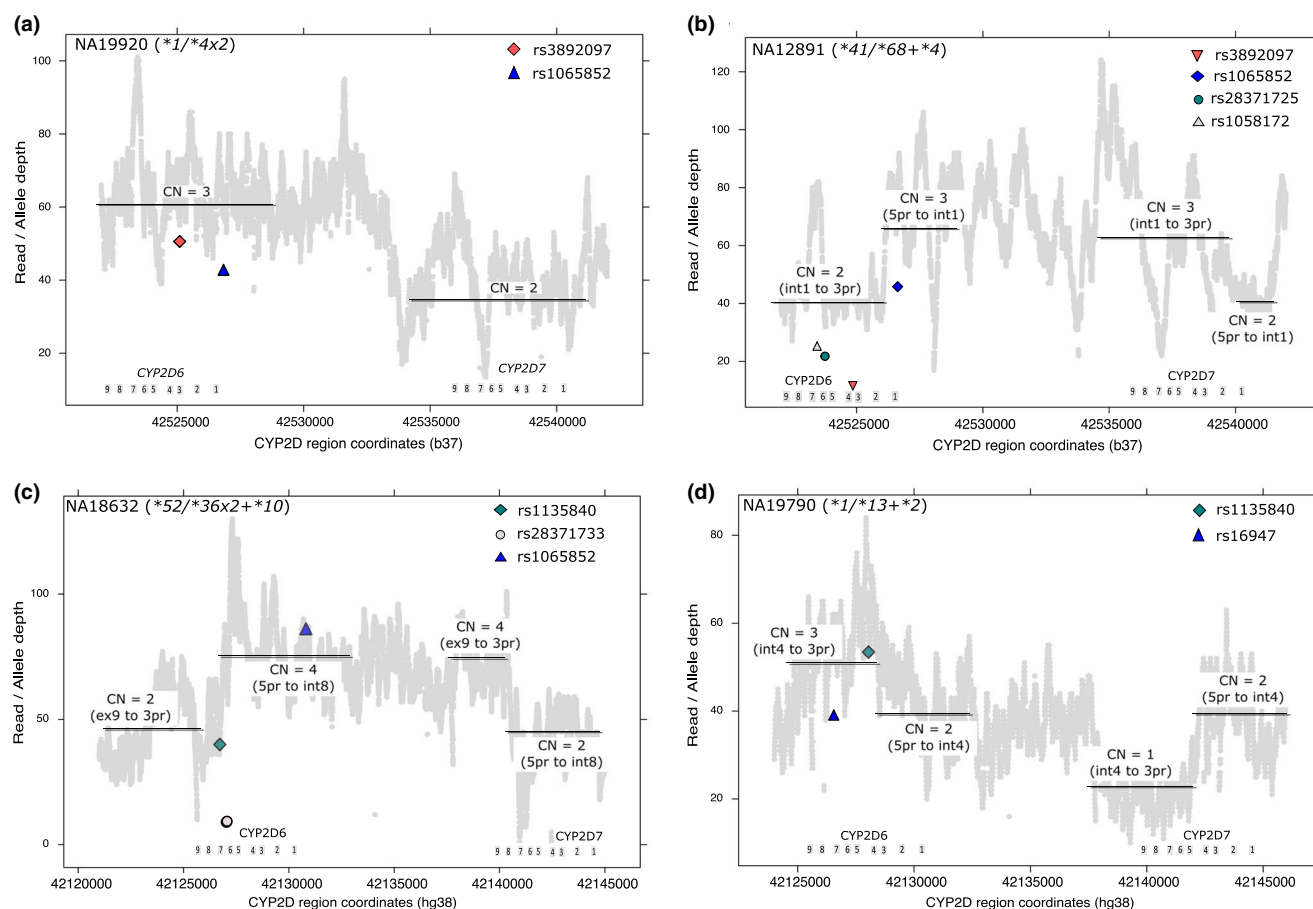


Figure 2 Examples of *CYP2D6* structural variants detected by StellarPGx in the WGS validation data set corresponding to the GeT-RM reference materials. (a) Shows the *CYP2D6* region coverage in an admixed American sample NA19920 with the *CYP2D6**1/*4x2 diplotype. Note the higher *CYP2D6* copy number (CN = 3) compared with *CYP2D7* (CN = 2) and the comparable allele depth for the core variants (rs3892097 and rs1065852). (b) Shows a European sample NA12891 with the *CYP2D6**41/*68+*4 diplotype. Note the hybrid read distribution due to the *68 allele and the intron 1 switch region. Also, the allele depth for rs1065852 is much higher than that for rs3892097. (c) Shows an East Asian sample NA18632 with the complex *CYP2D6**52/*36x2+*10 diplotype. Note the exon 9 conversion showed by diminished coverage for the *CYP2D6* exon 9 and elevated coverage for *CYP2D7* exon 9. (d) Shows sample NA19790 with the complex *CYP2D6**1/*13+*2 diplotype. Note the intron 4 switch region for the *CYP2D6**13 hybrid allele. CYP, cytochrome P450; ex, exon; int, intron; WGS, whole genome sequence; 3pr, 3-prime region; 5pr, 5-prime region.

targeted genotype panel results only,³⁸ thus probably missing the *106 defining variant (3878G>A), which appears in the WGS data. Therefore, we considered *CYP2D6**29/*106 as the truth call for NA18519 as did the Cyrius authors.²⁵ The assays used to generate the GeT-RM truth calls are described in detail by Gaedigk *et al.*³⁷

StellarPGx's *CYP2D6* star allele calls were highly concordant (99%) with the GeT-RM reference materials and comparable to Cyrius calls (Table 1). Notably, StellarPGx was able to correctly genotype sample NA19908 (*CYP2D6**1/*46—normal metabolizer phenotype) as the pipeline queried multiple sub-allele combinations, unlike Cyrius which ambiguously reported *CYP2D6**1/*46 or *1/*43 (indeterminate phenotype) for this sample. In comparison, Aldy and Stargazer diplotype calls had lower concordance with GeT-RM calls (Table 1).

Furthermore, StellarPGx's phenotype assignments were 100% concordant with those corresponding to the GeT-RM reference materials. Cyrius had comparable phenotype concordance (99%), while Aldy (92%) and Stargazer (90%) had lower phenotype

assignment accuracy. Activity scores and metabolizer phenotypes were assigned according to consensus recommendations by Caudle *et al.*⁶ Detailed diplotype calls and corresponding phenotype assignments for each sample by each tool are provided in Table S3.

Sample NA18545 (*CYP2D6**5/*36x2+*10x2) was particularly challenging for all four tools as the *5 allele could not be detected in the presence of the *CYP2D6**36x2+*10x2 allele on the other chromosome. This sample was not genotyped by Stargazer while StellarPGx, Cyrius, and Aldy genotyped it as *CYP2D6**36+*10/*36+*10. Both diplotypes, however, translate to an intermediate metabolizer phenotype when applying the Clinical Pharmacogenetics Implementation Consortium (CPIC)—recommended translation method 6. StellarPGx consistently called the other diplotypes comprising *CYP2D6* hybrid rearrangements (Figure 2).

Notably, some indel positions in complex gene regions had to be shifted in our allele databases to call alleles such as *CYP2D6**40 and *CYP2D6**21. This shift in indel position is also implemented by the other allele callers.

Table 1 Summary of the CYP2D6 diplotype concordance for StellarPGx in comparison with other tools

Allele caller	Concordant samples	Concordance (%)
StellarPGx	108	99
Cyrius	107	98
Stargazer	92	84
Aldy	89	82

Summary of the CYP2D6 diplotype concordance for StellarPGx in comparison with other tools with respect to the 109 GeT-RM WGS samples used in the study.

CYP, cytochrome P450; GeT-RM, Genetic Testing Reference Material Coordination Program; WGS, whole genome sequence.

StellarPGx's performance on a larger CYP2D6 diplotype sample space

Given that the number of CYP2D6 alleles represented in the GeT-RM reference materials was rather limiting, we further tested the ability of StellarPGx's combinatorial diplotype assignment to resolve 35,778 diplotype simulations. This includes all combinations of 267 SNV-defined haplotypes covering CYP2D6*1.001*140.001 (see Table S4) from PharmVar v4.2.4 corresponding to GRCh37. StellarPGx correctly resolved 35,764 diplotypes and reported ambiguous calls for 14 cases. Examples of the ambiguous cases encountered include CYP2D6*132.001/*9.001 (called as CYP2D6*10/*115 or *132/*9), CYP2D6*1.001/*6.003 (called as CYP2D6*1/*6 or *39/*6), and CYP2D6*2.004/*6.002 (called as CYP2D6*2/*6 or *34/*6). See Supplementary Materials S1 and Table S4 for more details on the simulation analysis.

StellarPGx's performance on 11 additional CYP genes

StellarPGx's concordance with GeT-RM calls for CYP2A6, CYP2B6, CYP2C19, CYP2C9, CYP2C8, CYP3A4, CYP3A5, CYP2A1, CYP1A2, CYP2E1, and CYP4F2 diplotypes is summarized in Table 2. Briefly, StellarPGx had 100% concordance with GeT-RM diplotype calls 48 for CYP1A2, CYP3A4, CYP3A5, and two CYP2C cluster genes, i.e., CYP2C9 and CYP2C8. We also obtained highly comparable diplotypes with GeT-RM calls for CYP4F2 (96%), CYP2C19 (97%), and CYP2A6 (88%). On the other hand, genotyping CYP1A1, CYP2E1, and CYP2B6 was more challenging as the StellarPGx concordance with GeT-RM calls with respect to these genes was 78%, 63%, and 66%, respectively. For a detailed explanation of the discrepancies in the called diplotypes for CYP1A1, CYP2E1, and CYP2B6, see Discussion section. Detailed star allele calls for each gene/sample are provided in Table S5.

Of note, we identified samples with possible novel alleles for some CYP genes. For example, samples NA07056, NA07029, and NA18564 were genotyped as CYP4F2*3/*3 by GeT-RM, but StellarPGx flagged that a possible novel CYP4F2 allele was present as rs2108622 (CYP4F2*3) and rs3093105 (CYP4F2*2) were deduced to be on the same chromosome in each of these samples.

Run time comparisons

We compared the run time of StellarPGx against Aldy, Stargazer, and Cyrius for CYP2D6 star allele calling on a cluster of 4

computers, each with dual Intel Silver 4114 processors and 132GB of RAM, and running CentOS 7.5 (<https://www.centos.org>). Each task could use 20 GB memory and 10 parallel threads. For a single WGS sample run, Aldy was the fastest tool with a run time of 7 seconds while StellarPGx (58 seconds), Stargazer (91 seconds), and Cyrius (140 seconds) were significantly slower. However, for a multi-sample run (75 WGS samples aligned to build 37), StellarPGx had the fastest run time (4 minutes 7 seconds) followed by Aldy (6 minutes) and Stargazer (9 minutes 51 seconds), while Cyrius was significantly slower (57 minutes 49 seconds when using a single manifest file). StellarPGx was able to obtain the fastest run time for the multisample run owing to the efficient parallelization facilitated by its Nextflow implementation.

DISCUSSION

Accurately calling star alleles in CYP genes is integral to genotype-guided therapy for a wide array of drugs, and characterizing pharmacogenomic variation at the population level. Reproducible bioinformatics workflows are crucial in tackling the challenging task of genotyping this class of pharmacogenes using targeted and/or WGS data from the widely used short-read NGS technologies. Existing tools, notably Astrolabe, Aldy, Stargazer, and Cyrius have provided short-read-based genotyping solutions for CYP and/or other pharmacogenes in recent years. However, more tools with diverse approaches are still needed especially to support high confidence ensemble CYP genotyping that we propose in our previous benchmarking study.³⁸ As long read technologies such as PacBio^{13,14} and Oxford Nanopore^{15,16} become more widely used, adapting star allele callers to the analysis of targeted and/or WGS long-read data will presumably be easier, but nonetheless important.

We have described StellarPGx, a Nextflow pipeline for determining diplotypes in CYP genes from short-read HTS data, and

Table 2 Summary of the diplotype concordance for StellarPGx with respect to 11 additional CYP genes

Gene	Number of samples	Concordant samples	Concordance (%)
CYP2C8	74	74	100
CYP2C9	74	74	100
CYP3A4	74	74	100
CYP3A5	74	74	100
CYP1A2	67	67	100
CYP2C19	74	72	97
CYP4F2	67	64	96
CYP2A6	67	59	88
CYP1A1	67	52	78
CYP2B6	74	49	66
CYP2E1	67	42	63

Summary of the diplotype concordance for StellarPGx with respect to 11 additional CYP genes in comparison with the GeT-RM reference materials. The sample number ranges from 67 to 75 (all aligned to build 37) as not all the 109 aforementioned WGS samples had publicly available truth calls from GeT-RM.

CYP, cytochrome P450; GeT-RM, Genetic Testing Reference Material Coordination Program; WGS, whole genome sequence.

demonstrated its analytic accuracy using *CYP2D6* as a model CYP gene. StellarPGx uses a hybrid approach involving genome graph-based variant calling, read coverage information from the original linear alignments, and combinatorial star allele assignment. StellarPGx's *CYP2D6* allele calling performance is comparable to the recently developed Cyrius pipeline, and is superior to Aldy and Stargazer based on the analysis of WGS data corresponding to the GeT-RM reference materials.

Furthermore, StellarPGx's phenotype assignments were more concordant with the GeT-RM reference materials compared with any of the other allele callers (**Table S3**). The effect of genotyping accuracy on phenotype assignment, however, depends on the sample diplotype. For example, sample NA18565 (*CYP2D6**10/*36×2) was correctly assigned an intermediate metabolizer phenotype by all four allele callers, even though Aldy and Stargazer incorrectly called it as *CYP2D6**10/*36+*10. Conversely, sample NA18524 (*CYP2D6**1/*36×2+*10, normal metabolizer) was incorrectly assigned an intermediate metabolizer phenotype by Stargazer, which genotyped it as (*CYP2D6**36 + 10/*36+*10).

Though comparable in genotyping performance, StellarPGx offers notable advantages over Cyrius. Firstly, StellarPGx is able to call diplotypes in up to 12 CYP genes while Cyrius is currently only available for *CYP2D6*. Secondly, owing to its implementation in Nextflow and containerization with both Docker and Singularity, StellarPGx installs easily across various computational platforms and affords high reproducibility and scalability. In the same vein, StellarPGx calls star alleles in a shorter time compared with Cyrius especially for large cohorts. In addition, users can easily perform modifications to the StellarPGx pipeline by including custom Nextflow processes for performing different steps. One such modification would be to run a tool like Aldy in a parallel process so as to obtain consensus genotype calls for the same CYP gene(s) in multiple samples. Furthermore, the use of pangenome-based variant calling in the initial process of the StellarPGx pipeline enhances allele-calling confidence. In addition, StellarPGx is able to detect possible occurrences of novel alleles by flagging novel combinations of existing core SNVs and also interrogating whether novel core SNVs (especially missense, frameshift, stop loss, and stop gain variants) are present in a sample.

As shown in our simulations, StellarPGx is currently not able to resolve diplotypes with exactly similar variant combinations (see **Supplementary Materials S1**), thus resulting in ambiguous diplotype assignments for such cases. This limitation is also common to other allele callers including Astrolabe,²⁰ Aldy,²¹ and Cyrius.²⁵ Future updates to the StellarPGx database with alleles upgraded from limited to moderate or definitive level of evidence by PharmVar will likely mitigate this challenge. Another main limitation is having to perform structural variant detection by complementing GraphTyper's structural variant genotypes with coverage information from original linear alignments. Even though genome graph-based variant detection has proved to be excellent for genotyping SNVs, full gene deletion, and full gene duplications, we observed a number of false positive and false negative genotypes for the hybrid rearrangements in the model gene (*CYP2D6*) and challenges in including them in the pangenome construction, hence resorting to the hybrid allele calling approach. Additionally,

the realignment by GraphTyper to the pangenome is done for reads already aligned to a genomic region of interest in the original linear reference alignment, which could introduce some reference-based alignment bias.

Further, StellarPGx's diplotype concordance was lower for *CYP1A1*, *CYP2E1*, and *CYP2B6* compared with other supported CYP genes. For *CYP2B6* in particular, StellarPGx had a high number of ambiguous diplotype calls (20 samples) with respect to the data set used for our validation. For example, samples HG00436, NA06991, NA07029, NA07055, and NA10847 (genotyped as *CYP2B6**1/*6 by GeT-RM as *6 is considerably more frequent than *4) were genotyped as *CYP2B6**1/*6 or *4/*9 by StellarPGx since both diplotypes are equally likely given the allele defining variant combinations in the samples. For *CYP1A1* and *CYP2E1*, one possible explanation for the inconsistent diplotype calls is that the number of variants tested by some of the genotyping panels used by GeT-RM was rather limiting. For example, sample NA18980 was called as *CYP1A1**1/*2 by GeT-RM, while StellarPGx called it as *CYP1A1**13/*2B since we were able to detect the *CYP1A1**13 defining variant rs4646422. Notably, for *CYP2E1*, only two panels were used for generating consensus reference *CYP2E1* genotypes by GeT-RM. Indeed most of the discrepant *CYP2E1* diplotypes between StellarPGx and GeT-RM arose in samples that did not have definitive consensus calls by GeT-RM. For example, samples HG00436 and NA18980 were homozygous for both *CYP2E1**5B (rs3813867, rs2031920) and *CYP2E1**7 (rs2070673) defining variants. GeT-RM genotyped them as *CYP2E1*(*7/*7), indicating uncertainty in the final diplotype, while StellarPGx had "no calls" for these samples and indicated possible presence of novel alleles. Similarly, sample NA18992 was genotyped by GeT-RM as *CYP2E1*(*7)/*7 while StellarPGx detected the potentially function-altering rs199856651 in this sample. Furthermore, possible novel alleles were also identified for *CYP4F2* and *CYP2A6* in the validation data set (see **Table S5**). These findings indicate that more comprehensively characterized reference materials are needed for all core CYP genes as has been undertaken for *CYP2D6* recently.³⁷ Additionally, updates to PharmVar's star allele definitions for genes such as *CYP2E1* and *CYP2A6* will allow for allele definitions to better reflect variation in their loci.

Future developments of StellarPGx include adding support for more CYP genes as well as related pharmacogenes such as *POR*. We also intend to make StellarPGx a fully pangenome-based star allele calling pipeline as structural variant genotyping using genome graph approaches improves. Additionally, adding support for long-read sequence data and targeted NGS data from panels such as PGRNseq⁴⁷ would enhance the utility of StellarPGx. However, analyzing whole exome sequence data is still a major challenge as intronic SNVs and copy number change-points are also important for StellarPGx's star allele calling.

For short-read data, we recommend running StellarPGx using high coverage (at least 25×) WGS data especially when genotyping *CYP2D6*, *CYP2A6*, *CYP2B6*, and *CYP2E1*. Further efforts aim to keep the automated phenotype prediction up to date as activity score systems and phenotype assignment recommendations are refined by clinical pharmacogenomics consortia.

We plan to continue enhancing StellarPGx's ability to detect novel alleles as well as complex population-specific alleles as community usage increases, especially from studies using NGS data from underrepresented populations. StellarPGx's allele databases will also be updated on a regular basis to match the updates to reference databases such as PharmVar 10, especially as comprehensive allele characterization of genes such as *CYP2E1* and *CYP2A6* becomes available.

We recommend running StellarPGx in a multinode cluster-environment using the prebuilt Singularity image. However, for single sample cases, the pipeline can also be run on a personal computer (MacOS or Linux) preferably enhanced with external disk storage. StellarPGx is freely available from <https://github.com/SBIMB/StellarPGx> under the MIT License.

SUPPORTING INFORMATION

Supplementary information accompanies this paper on the *Clinical Pharmacology & Therapeutics* website (www.cpt-journal.com).

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CONFLICT OF INTEREST

The authors declared no competing interests for this work.

AUTHOR CONTRIBUTIONS

D.T., B.I.D., G.E.B.W., A.S., J.d.R., Z.L., and S.H. wrote the manuscript. D.T., B.I.D., G.E.B.W., J.d.R., Z.L., and S.H. designed the research. D.T. and S.H. performed the research. D.T. analyzed the data.

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